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Question Paper Code: 111804

MOUNT ZION COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

B.E., / B.Tech., DEGREE EXAMINATIONS, NOVEMBER/DECEMBER - 2025

First Semester

CY25C01 - APPLIED CHEMISTRY -I

(Common to All Departments)

(Regulation 2025)

Time: 3 Hours

Maximum Marks: 100 Marks

Answer ALL Questions

PART A - (10 x 2 = 20 Marks)

1. Enumerate the hardness and its types.
2. State the requirements for industrial feed water.
3. What do you mean by nano-clusters?
4. List out various types of nanomaterials.
5. Define the term standard electrode potential.
6. Recall redox reaction with an example.
7. What do you mean by galvanic corrosion?
8. List out any two importance of galvanic series.
9. What are batteries ? Give its types.
10. Recall whether the output voltage of battery can be increased by its size or not?

PART B - (5 x 16 = 80 Marks)

11. (a) Explain in detail about the water quality parameters and standards. (16)

Or

- (b) Describe the steps involved in the municipal water treatment. (16)

12. (a) Demonstrate the size dependent properties and applications of nanomaterials. (16)

Or

- (b) Explain any two methods of preparation of nanomaterials. (16)

13. (a) Derive the Nernst equation for a single electrode potential and explain its advantages. (16)

Or

- (b) Evaluate the factors influencing the conductivity of electrolytes in detail. (16)

14. (a) Summarize the organic and inorganic coating methods with proper examples. (16)

Or

- (b) Determine the factors influencing the chemical and electrochemical corrosion. (16)

15. (a) With suitable diagram explain any one primary and secondary batteries. (16)

Or

- (b) Illustrate in detail about the battery materials, performance parameters, testing and safety aspects of batteries. (16)